

Tank sloshing natural-period & resonance explorer

Fundamental sloshing natural period across tank shapes — rectangular, upright cylinder, horizontal cylinder (road tanker) and elliptical/obround fuel tanks — as a normalized master curve that collapses every size onto one dimensionless family, a shape-selector lookup for real periods, and a resonance screen showing partial-fill coupling with vessel roll. Backed by a worldwide relationship survey.

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tank shapes

$$\Omega_1 = \omega_1 \sqrt{Lc/g}$$

normalized master curve

<0.5%

API 650 ↔ potential-flow

WHAT YOU GET

- Normalized dimensionless master curve (Ω_1 vs fill/slenderness ratio) — any size reads off one family
- Shape dropdown → real natural periods for rectangular, upright/horizontal cylinder and oval cross-sections (equivalent-rectangle for the tanker shapes)
- Resonance screen: partial fill sweeps the sloshing period into the vessel roll band — the coupling external diffraction (AQWA/OrcaWave) misses, and the analytical→CFD tiering behind the ballast-tank study
- Relationship basis surveyed worldwide: Abramson, Ibrahim, Faltinsen & Timokha, Mclver, Housner/API 650, plus road/automotive-tanker (UMTRI, Rajagounder, Micheli)